PROPERTIES OF RELATIONS

Relations and Digraphs (Ch4.4)

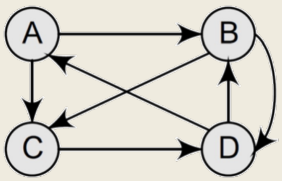
Roster form

{ (A,B), (A,C),
(B,C), (B,D),
(C,D), (D,A), (D,B) }

Matrices

$$\begin{array}{c|cccc} & A & B & C & D \\ \hline A & 0 & 1 & 1 & 0 \\ B & 0 & 0 & 1 & 1 \\ C & 0 & 0 & 0 & 1 \\ D & 1 & 1 & 0 & 0 \end{array}$$

Digraphs



Properties of Relations

- ❑ Reflexive
- ❑ Symmetric
- ❑ Asymmetric
- ❑ Irreflexive
- ❑ Anti-symmetric
- ❑ Transitive

Equivalence relations

Reflexive, symmetric, and transitive

Reflexive relation

R on A ; R is **reflexive** if $(a,a) \in R$ for every $a \in A$.

Every node has the pair (a,a)

If A = {a,b} then R is **reflexive** if:

Roster Method	R = {(a,a),(b,b)}
Directed Graph	Every vertex has a self-loop *has a cycle of length 1 at all vertices*
Incidence Matrix	Main diagonal all "1s" $\begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix}$

https://calcworkshop.com/relations/discrete-math-relations/

WORKED EXAMPLES

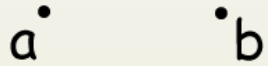
Determine the properties of the given relations on A = {1, 2, 3}. If you say a relation does not have one of the properties, give a counterexample showing why the relation does not have it.	Reflexive
$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$	
R = {(1,1), (1,2), (2,1), (2,2), (2,3), (3,3)}	

Irreflexive relation

R on A ; R is **irreflexive** if $(a,a) \notin R$ for every $a \in A$.

No nodes have the pair (a,a)

If set A = {a,b} then R is **irreflexive** if:

Roster Method	$(a,a) \notin R$ *and* $(b,b) \notin R$
Directed Graph	Every vertex does <u>not</u> have a self-loop *has no cycle of length 1 at any vertices* 
Incidence Matrix	Main diagonal all "0s" $\begin{bmatrix} 0 & & \\ & 0 & \\ & & 0 \end{bmatrix}$

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WORKED EXAMPLES

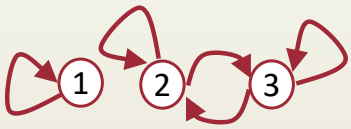
Determine the properties of the given relations on A = {1, 2, 3}. If you say a relation does not have one of the properties, give a counterexample showing why the relation does not have it.	Irreflexive
$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$	
R = {(1,1), (1,2), (2,1), (2,2), (2,3), (3,3)}	

Reflexive vs. Irreflexive Relations

Not Reflexive ≠ Irreflexive	Not Irreflexive ≠ Reflexive

- A non-reflexive relation may or may not be irreflexive
- A non-irreflexive relation may or may not be reflexive

- ❑ Reflexive
- ❑ Irreflexive



- ❑ Reflexive
- ❑ Irreflexive

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

- ❑ Reflexive
- ❑ Irreflexive

The equality relation

Δ is the relation of equality on the set A, aka the diagonal relation

$\Delta = \{(\mathbf{a}, \mathbf{a}) \mid \mathbf{a} \in \mathbf{A}\}$

Δ is reflexive

Try it out

True or false: a relation R on A is reflexive iff $\Delta \subseteq R$

True or false: a relation R on A is irreflexive iff $\Delta \cap R = \phi$

Symmetric relation

If set A = {a,b} then R is **symmetric** if:

(a,b) in R → (b,a) in R

(a,b) not in R → (b,a) not in R

Roster Method	$(a,b) \in R$ *and* $(b,a) \in R$
Directed Graph	Every pair of vertices is connected by none or exactly two directed lines in opposite directions A vertex may or may not have a self-loop
Incidence Matrix	$m_{ij} = m_{ji}$ off the main diagonal main diagonal *don't care*

$$\begin{bmatrix} & 1 & \\ 1 & \ddots & 0 \\ & 0 & \end{bmatrix}$$

Antisymmetric relation

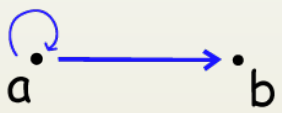
If set A = {a,b} then R is **antisymmetric** if:

WHEN a≠b THEN

(a,b) not in R OR

(b,a) not in R

Roster Method	*if* $(a,b) \in R$ *and* $(b,a) \in R$ *then* $a = b$ *if* $a \neq b$, *then either* $(a,b) \notin R$ *or* $(b,a) \notin R$.
Directed Graph	Every pair of vertices is connected by none or exactly one directed lines A vertex may or may not have a self-loop
Incidence Matrix	$m_{ij} \neq m_{ji}$ off the main diagonal main diagonal *don't care*



$$\begin{bmatrix} & 0 & & \\ 1 & \ddots & 0 & \\ & 0 & \ddots & 1 \\ & & 0 & \end{bmatrix}$$

Caution Antisymmetric is not the logical negation of symmetric.

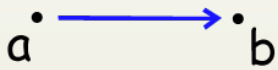
WORKED EXAMPLES

Determine the properties of the given relations on A = {1, 2, 3}. If you say a relation does not have one of the properties, give a counterexample showing why the relation does not have it.	Symmetric	Antisym
$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$		
R = {(1,1), (1,2), (2,1), (2,2), (2,3), (3,3)}		

Asymmetric relation

*A relation is **asymmetric** iff it's both antisymmetric & irreflexive*

If set A = {a,b} then R is **asymmetric** if:

Roster Method	*If whenever* $(a,b) \in R$, *then* $(b,a) \notin R$
Directed Graph	Every pair of vertices is connected by none or exactly one directed lines Every vertex does not have a self-loop 
Incidence Matrix	$m_{ij} \neq m_{ji}$ off the main diagonal main diagonal all "0"s $\begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ & 0 & 1 & \\ & & 0 & \end{bmatrix} \begin{bmatrix} 0 & & \\ & 0 & \\ & & 0 \end{bmatrix}$

Caution Asymmetric is not the logical negation of symmetric.

https://calcworkshop.com/relations/discrete-math-relations/

WORKED EXAMPLES

Determine the properties of the given relations on A = {1, 2, 3}. If you say a relation does not have one of the properties, give a counterexample showing why the relation does not have it.	Irreflexive	Antisym	Asym
$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$			
R = {(1,1), (1,2), (2,1), (2,2), (2,3), (3,3)}			

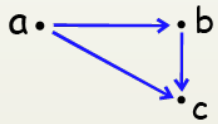
PRACTICE PROBLEMS

- True or false: a relation R on A is asymmetric iff $\triangle \cap R = \phi$

Transitive relation

If set A = {a,b} then R is **transitive** if:

if a and c are connected by a path of length 2 in R, aR^2c then they must also be connected by a path of length 1, aRc

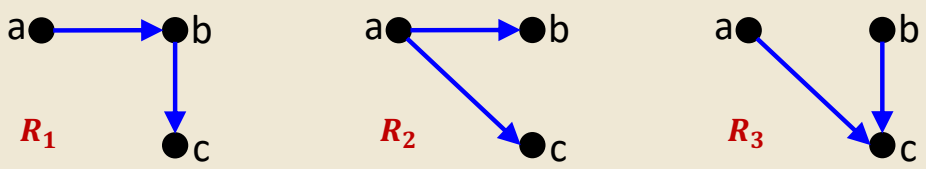
Roster Method	*if* $(a,b) \in R$ *and* $(b,c) \in R$ *then* $(a,c) \in R$
Directed Graph	❑ For every path joining three vertices a, b, and c, there is also a directed line joining a to c  ❑ On a digraph, R on A is transitive if whenever $a\ R^2\ c$, then $a\ R\ c$ ❑ If vertices a and c are connected by a path of length 2 in R, then a and c must also be connected by a path of length 1
Incidence Matrix	If $M_{R^2} = (M_R)^2_{\odot} = M_R \odot M_R = M_R$, then R is transitive

https://calcworkshop.com/relations/discrete-math-relations/

Transitive relation

- A relation R on A is transitive if $(a,b) \in R$ and $(b,c) \in R$ then $(a,c) \in R$

- What if there are no pairs (a,b) or (b,c) in R?
- Are the following relations transitive?



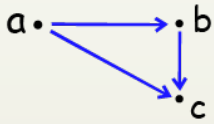
WORKED EXAMPLES

Determine the properties of the given relations on A = {1, 2, 3}. If you say a relation does not have one of the properties, give a counterexample showing why the relation does not have it.	Transitive
$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$	
R = {(1,1), (1,2), (2,1), (2,2), (2,3), (3,3)}	

Matrices of transitive relations

- A relation R is transitive iff $\boldsymbol{M_R} = [\boldsymbol{m_{ij}}]$ has the following property:

  if $m_{ij} = 1$ and $m_{jk} = 1$, then $m_{ik} = 1$



- If $M_{R^2} = (M_R)^2_{\odot} = M_R \odot M_R = M_R$, then R is transitive.

The converse is not always true, that is,
if R is transitive, then M_{R^2} need not be equal to M_R

WORKED EXAMPLES

Show that the relation is transitive by computing $M_{R^2} = M_R$

Earlier, we verify that $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ is transitive. Here, show using Boolean product.

$M_{R^2} = M_R \odot M_R = M_R \rightarrow R$ **is transitive**

PRACTICE PROBLEMS

Show that when $M_{R^2} \neq M_R$, R may or may not be transitive

Let $A = \{a, b, c, d\}$ and let R be a relation of A, $R = \{(a, b), (a, c), (d, b)\}$

Step1: compute M_{R^2} and M_R and show that they are not equal.

$M_R \odot M_R \neq M_R \not\rightarrow R$ **is not transitive**

PRACTICE PROBLEMS

Show that when $M_{R^2} \neq M_R$, R may or may not be transitive

Let $A = \{a, b, c, d\}$ and let R be a relation of A, $R = \{(a, b), (a, c), (d, b)\}$
Step2: verify that R is transitive

$M_R \odot M_R \neq M_R \not\rightarrow R$ **is not transitive**

EQUIVALENCE RELATIONS

Relations and Digraphs (Ch4.5)

Equivalence relation

A relation R on a set A is called an **equivalence relation** if it is

reflexive

symmetric

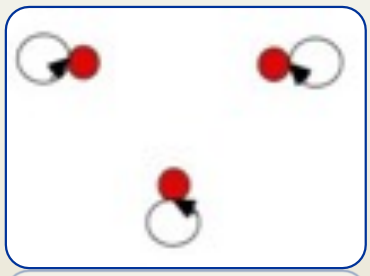
transitive

WORKED EXAMPLES

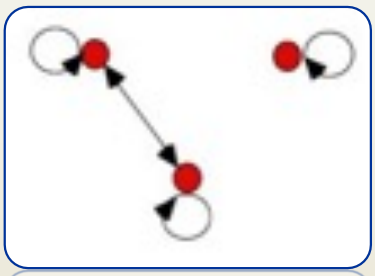
Determine the properties of the given relations on A = {1, 2, 3}. If you say a relation does not have one of the properties, give a counterexample showing why the relation does not have it.	Reflexive	Symmetric	Transitive	Equivalence
$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$				
$R = \{(1,1), (1,2), (2,1), (2,2), (2,3), (3,3)\}$				

Recognizing equivalence relations on digraphs is easy

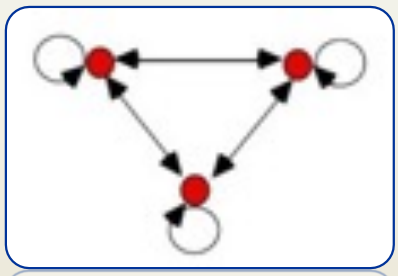
Each subset of nodes form a universal relation, containing all possible edges. Each sub(digraph) is a complete (sub)digraph. All edges are present.



3 nodes, 3 (sub)digraphs



3 nodes, 2 (sub)digraphs

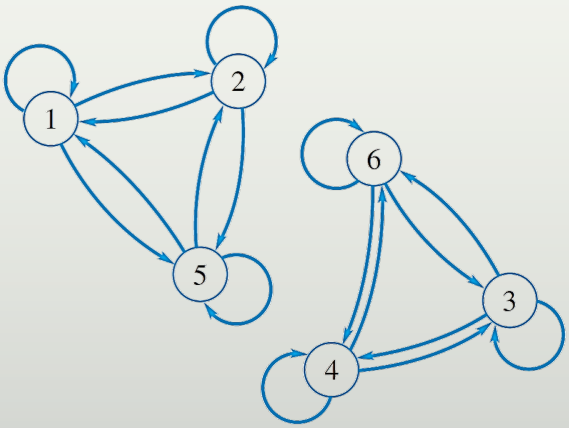


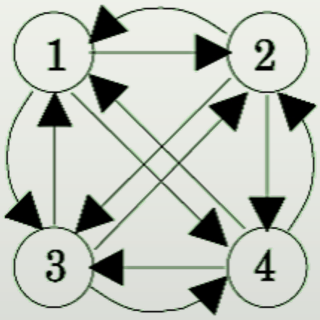
3 nodes, 1 (sub)digraphs

https://www.slideserve.com/vida/8-5-equivalence-relations

WORKED EXAMPLES

Given the digraphs of two relations. Are they equivalence relations?





Practice Problems on Properties of Relations

PRACTICE PROBLEMS

Determine the properties of the given relations. If you say a relation does not have one of the properties, give a counterexample showing why the relation does not have it.	Reflexive	Irreflexive	Symmetric	Antisym	Asym	Transitive
A = {a,b,c} R = {(a,a), (a,b), (b,b), (b,c), (c,a), (c,b), (c,c)}						
A = {a,b,c,d} R = {(a,b), (b,c), (c,a), (c,d), (d,a), (d,b)}						
A = {a,b,c,d,e} R = {(a,a), (b,a), (c,a), (d,a), (d,b), (d,c), (d,d)}						
A = {a,b,c,d,e} R = {(a,b), (b,c), (c,a), (d,d), (d,e), (e,d), (e,e)}						

PRACTICE PROBLEMS

Determine which relations have the following properties.

$$M_{R_1} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix} \qquad M_{R_2} = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$M_{R_3} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \qquad M_{R_4} = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

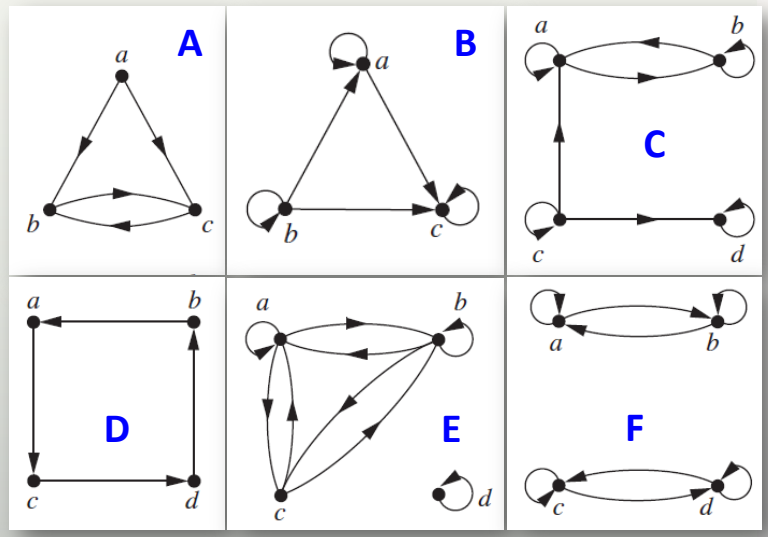
$$M_{R_5} = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad M_{R_6} = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

- Reflexive:
- Irreflexive:
- Symmetric:
- Antisymmetric:
- Asymmetric:
- Transitive:

PRACTICE PROBLEMS

- Which of the following digraphs are reflexive, irreflexive, symmetric, antisymmetric, asymmetric, or transitive?

	A	B	C	D	E	F
Reflexive						
Irreflexive						
Symmetric						
Anti-symmetric						
Asymmetric						
Transitive						



PRACTICE PROBLEMS

Determine the properties of the given relations. If you say a relation does not have one of the properties, give a counterexample showing why the relation does not have it.	Reflexive	Irreflexive	Symmetric	Antisym	Asym	Transitive	Equivalence
$R = \{(a,b) \in \mathbb{Z}\times\mathbb{Z} \mid a = b^2 + 1\}$							
$A = \mathbb{Z}^+$, $R = \{(a,b) \in A\times A \mid a \text{ divides } b\}$							
$R = \{(a,b) \in \mathbb{R}\times\mathbb{R} \mid 1 \leq \lvert a\rvert + b \leq 2\}$							
R on $\mathbb{R}$; $x\ R\ y$ iff $x - y + \sqrt{2}$ is an irrational number *Irrational numbers cannot be expressed as the ratio of two integers.*							

PRACTICE PROBLEMS

Determine the properties of the given relations. If you say a relation does not have one of the properties, give a counterexample showing why the relation does not have it.	Reflexive	Irreflexive	Symmetric	Antisym	Asym	Transitive	Equivalence
$A = \mathbb{Z}$, $R = \{(a,b) \in A\times A \mid a \equiv b \pmod 2\}$ This is equivalent to saying a is congruent to b mod 2, i.e. a and b yield the same remainder when divided by 2}							
A is the set of all triangles in the xy-plane, $R = \{ (a,b) \in A\times A \mid$ triangle a and triangle b have exactly the same three sides and the same three angles $\}$							